ArtBench
Generative Modeling

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Abstract. This paper presents a comparative study of generative models applied to the visual arts domain using the ArtBench-10 dataset. We implement, train, and evaluate three distinct families of generative architectures: Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), and Diffusion Models. Our objective is to synthesize high-fidelity, diverse, and stylistically consistent images of paintings while navigating the inherent trade-offs of each paradigm. To ensure statistical robustness, we evaluate the models quantitatively using Fréchet Inception Distance (FID) and Kernel Inception Distance (KID) averaged across multiple random seeds. Our final findings reveal that the Pixel-Space Diffusion model achieves the highest generative quality.

Keywords: Generative AI · Image Generation · ArtBench · Generative Models · GANs · Variational Autoencoders · Diffusion Models · FID · KID.

1 Introduction

Generative Artificial Intelligence (AI) refers to systems capable of creating novel content rather than merely analyzing or processing existing data [10]. As one of the most transformative technologies of recent years, it has seen remarkable advancements, particularly in the synthesis of high-fidelity images. The application of these models to the visual arts requires capturing complex distributions of brushstrokes, color palettes, and distinct artistic styles. This project, conducted as part of the Generative AI course at the University of Coimbra, aims to learn generative models from artwork images and produce plausible new samples matching the data distribution.

We explore three main families of generative models: Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), and Diffusion Models. This comparative study is motivated by the distinct theoretical foundations each family offers. Each architecture offers a distinct approach to generative tasks. While VAEs leverage probabilistic latent spaces for efficient encoding and GANs utilize adversarial training to achieve high visual fidelity, Diffusion models employ iterative refinement to deliver superior resolution and semantic consistency, effectively addressing the limitations of their predecessors [18].

By examining these diverse architectures, this project seeks to identify the optimal trade-off between computational efficiency and the quality required for

artistic generation. By building a complete pipeline from data loading to evaluation, we compare these approaches based on their visual fidelity, diversity, and style consistency.

2 Problem Description

The main objective is to design, train, and evaluate generative models. The target domain for this study is the ArtBench-10 dataset, a curated collection designed to benchmark generative models [9]. The dataset comprises 60,000 images across 10 distinct artistic styles, featuring a balanced distribution of 5,000 training images and 1,000 testing images per style, all at a 32x32 RGB resolution.

ArtBench-10 offers significant advantages over previous artwork datasets. Crucially, its class-balanced nature and high-quality, clean annotations are particularly beneficial for the rigorous evaluation required in this project.

The primary challenge lies in the nature of the generative task. Unlike discriminative tasks, such as classification, success cannot be measured by a simple accuracy metric. Instead, the models must balance visual fidelity with diversity and style consistency. This necessitates a comparative framework to determine how different architectures handle the latent representations of art. Consequently, the performance of each model will be assessed through rigorous quantitative distribution-level metrics, such as the Fréchet Inception Distance (FID), which remains a widely used benchmark for comparing generative image models [2].

3 Methodology and Approach

To comprehensively explore generative modeling, we implemented variants across the three required architectural families.

3.1 Variational Autoencoders (VAEs)

Variational Autoencoders (VAEs) are generative models introduced by [8] that address the problem of learning directed probabilistic models. They are autoencoders that learn a directed latent-variable model alongside approximate posterior inference [7]. These models have emerged as one of the most popular approaches to the unsupervised learning of complicated distributions [3].

Convolutional VAE After establishing baselines with deterministic Dense and Convolutional Autoencoders, we shifted our focus to generative architectures. The exploration of generative architectures began with the Convolutional VAE (ConvVAE), which integrates the standard VAE probabilistic framework with convolutional layers [14]. The encoder consists of three 2D Convolutional Layers (Conv2d), each followed by Batch Normalization (BatchNorm2d) and Rectified Linear Unit (ReLU) activation. The first two layers utilize a stride of 2 to reduce

spatial dimensions from 32×32 to 8×8, while the third maintains the resolution with a stride of 1. The resulting 8192-dimensional flattened feature map feeds into two parallel linear heads—fc_mu and fc_logvar—each outputting a 128-dimensional vector representing the mean and log-variance of the latent Gaussian distribution. Sampling is performed via the reparameterization trick:

$$z = \mu + \epsilon \cdot \exp\left(\frac{\log(\sigma^2)}{2}\right), \text{where} \epsilon \sim \mathcal{N}(0, I)$$

The decoder projects z back to a $128 \times 8 \times 8$ representation via a linear layer, followed by three ConvTranspose2d layers (channels: $128 \to 64 \to 32 \to 3$). The final two layers utilize stride-2 upsampling to recover the 32×32 resolution, concluding with a Sigmoid activation to constrain pixel outputs to the $[0, 1]$ range.

To further enhance capacity and capture fine-grained features, a deeper variant of the Convolutional VAE was implemented. The encoder is extended with two additional convolutional layers (3→32→64→128→256→512), and an extra stride-2 layer reduces the spatial map to 4×4 rather than 8×8. This yields a flattened dimension of 8192 (512×4×4), providing the model with substantially increased representational depth. The bottleneck and reparameterization mechanisms remain consistent with the base architecture. The decoder mirrors this deeper structure with five ConvTranspose2d layers (512→256→128→64→32→3), again terminating with a Sigmoid activation.

Conditional VAE The Conditional VAE (CondVAE) builds upon the base architecture with structural modifications to both the encoder and decoder to facilitate class-conditioned generation [15]. This model was implemented as an experimental framework to investigate the variance in reconstruction loss across different art styles.

Denoising VAE The Denoising VAE (dVAE) utilizes the same structural backbone as the base VAE but shifts the training paradigm to prioritize robustness. In this setup, the input data is intentionally corrupted with Gaussian noise before entering the encoder. The architecture is the same as that of the ConvVAE. The distinction lies in the training procedure: prior to each forward pass, Gaussian noise is injected into the input using

$$\tilde{\mathbf{x}} = \text{clip}(\mathbf{x} + \sigma \cdot \boldsymbol{\epsilon}, 0, 1)$$

where $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ represents the Gaussian noise and σ defines the noise intensity [6].

3.2 Generative Adversarial Networks (GANs)

Deep Convolutional GAN (DCGAN) Our baseline adversarial model follows the Deep Convolutional GAN (DCGAN) architecture, tailored for the $32 \times 32 \times 3$ spatial resolution of the ArtBench-10 dataset.

The Generator takes a noise vector $z \sim \mathcal{N}(0, I)$ from a 100-dimensional latent space and projects it through a series of fractional-strided convolutions (`ConvTranspose2d`). Each layer, except the output, is followed by Batch Normalization and a ReLU activation. The final layer utilizes a Tanh activation to map the generated images to a $[-1, 1]$ pixel range.

The Discriminator mirrors this structure using strided convolutions to downsample the image, paired with LeakyReLU activations (negative slope of 0.2). To address the notorious training instability and mode collapse common in standard GANs, we introduced several modern stabilization techniques into our DCGAN baseline:

- Spectral Normalization: We removed Batch Normalization from the Discriminator and instead applied Spectral Normalization to all convolutional weights. This strictly bounds the Lipschitz constant of the Discriminator, stabilizing its gradients during adversarial training.
- Hinge Loss: Instead of the standard min-max Binary Cross-Entropy loss, we optimized the network using the geometric Hinge Loss. This pushes the Discriminator outputs apart while preventing vanishing gradients for the Generator.
- Two Time-Scale Update Rule (TTUR): We decoupled the optimizers for the Generator and Discriminator, allowing them to use distinct learning rates (α_G and α_D). We also parameterized n_{critic} steps, enabling the Discriminator to update multiple times per Generator step.

Following standard practices for stable adversarial training, network weights for convolutional layers were initialized from a zero-centered Normal distribution ($\mu = 0, \sigma = 0.02$), while Batch Normalization weights were initialized with $\mu = 1.0$ and $\sigma = 0.02$ [12].

Wasserstein GAN with Gradient Penalty (WGAN-GP) While the DCGAN architecture provides a strong baseline, adversarial training using the Jensen-Shannon divergence often suffers from vanishing gradients and mode collapse. To address this, we implemented a Wasserstein GAN with Gradient Penalty (WGAN-GP) [4], which minimizes the Earth-Mover (Wasserstein-1) distance between the real and generated distributions.

The Generator retains a fractional-strided convolutional architecture similar to our DCGAN baseline. However, the Discriminator is replaced by a *Critic* network. Unlike a standard discriminator that outputs a probability in $[0, 1]$, the Critic outputs an unbounded scalar score representing the Wasserstein distance. To enforce the 1-Lipschitz continuity required by the Wasserstein metric, we applied a gradient penalty:

$$L_{GP} = \lambda \mathbb{E}_{\hat{x} \sim \mathbb{P}_{\hat{x}}} \left[(||\nabla_{\hat{x}} C(\hat{x})||_2 - 1)^2 \right]$$

where $\hat{x}$ are random linear interpolations between real and fake samples, and we set $\lambda = 10$.

Because the gradient penalty penalizes the norm of the critic's gradient with respect to each input independently, batch-level normalizations are invalid. Therefore, we replaced Batch Normalization in the Critic with Layer Normalization (`LayerNorm`), which normalizes features independently for each sample.

Additionally, to prevent the Critic's unbounded outputs from drifting arbitrarily far from zero during training, we incorporated a small drift penalty, $L_{drift} = \epsilon \mathbb{E}_x[C(x)^2]$, with $\epsilon = 0.001$.

The networks were optimized using Adam with momentum parameters $\beta_1 = 0.0$ and $\beta_2 = 0.99$. To ensure the Critic provides accurate Wasserstein distance estimates, we updated the Critic five times ($n_{critic} = 5$) for every single Generator update.

3.3 Diffusion Models

Diffusion models learn to synthesize data by reversing a gradual noising process. We implemented two distinct diffusion pipelines: a standard Pixel-Space Diffusion model and a Latent Diffusion Model (LDM).

Pixel-Space Diffusion Our baseline diffusion model operates directly on the $32 \times 32 \times 3$ image space. The reverse denoising network is parameterized as a U-Net [5]. To condition the network on the diffusion timestep t, we utilized Sinusoidal Positional Embeddings, which are projected and added to the feature maps inside residual blocks. The U-Net employs Group Normalization, SiLU activations, and incorporates Self-Attention blocks at lower spatial resolutions (16×16 and 8×8) to capture global contextual relationships.

The forward diffusion process utilizes a *cosine* variance schedule [11], which is specifically advantageous for low-resolution images as it prevents the premature destruction of signal compared to standard linear schedules. The network was trained to predict the added noise (ϵ-prediction) using the Mean Squared Error (MSE) objective, optimized via AdamW with gradient clipping.

To enhance generation quality and stabilize the reverse process, we maintained an Exponential Moving Average (EMA) of the model weights with a decay rate of 0.999. All evaluations and final inferences were conducted using the EMA weights. Furthermore, to accelerate generation, we employed the Denoising Diffusion Implicit Models (DDIM) [16] sampling strategy, reducing the reverse process from 1000 training timesteps to just 100 inference steps.

Latent Diffusion Model (LDM) Operating directly in pixel space can be computationally prohibitive. To address this, we implemented a Latent Diffusion Model (LDM) [13], which separates the generative process into two distinct stages: perceptual compression and latent diffusion.

For the compression stage, rather than utilizing our baseline autoencoders (which utilized flat latent vectors), we trained a dedicated, unconditional Convolutional VAE (ConvVAE) from scratch. This ConvVAE was specifically designed to project the $32 \times 32 \times 3$ RGB images into a compact $8 \times 8 \times 4$ spatial latent representation. To ensure the latent space was highly expressive and structurally robust for diffusion, we implemented several architectural refinements: we utilized 4 latent channels to prevent the loss of critical high-frequency image details during compression, and replaced Batch Normalization with Group Normalization across both the ConvVAE and the diffusion network to maintain stability independent of batch size dependencies.

A critical mathematical requirement for standard diffusion noise schedules is that the target data distribution maintains roughly unit variance ($\sigma \approx 1$). Because the raw latent space of a VAE does not naturally conform to this, a mismatch with the cosine noise schedule can occur. To rectify this, we introduced a latent scaling factor ($s \approx 0.18215$). During training, the encoded latents are multiplied by this factor before diffusion; during inference, the generated latents are appropriately unscaled before being passed to the ConvVAE decoder.

A specialized `LatentUNet` was then trained entirely within this scaled latent space. To handle the complex transitions of the latent manifold, we increased the depth of the network by adding additional ResNet blocks. The model was optimized using AdamW with weight decay to provide better regularization. During inference, noise is iteratively denoised in the latent space using the DDIM sampler, and the final latent tensor is passed through the frozen ConvVAE decoder to recover the 32×32 image. This approach significantly reduces the spatial dimensionality of the diffusion process while maintaining high-fidelity perceptual details.

4 Experimental Setup

Our experimental pipeline was designed to ensure rigorous hyperparameter selection, efficient compute utilization, and statistically robust final evaluations.

4.1 Training Strategy and Hyperparameter Tuning

Following the project guidelines, the development phase was conducted on a 20% subset (10,000 images) of the ArtBench-10 training data to facilitate rapid iteration. Within this environment, we employed Optuna, a Bayesian hyperparameter optimization framework, to systematically refine our models. However, our tuning strategy was explicitly adapted to the distinct training dynamics of each architectural family.

For the Variational Autoencoders (VAEs), which possess highly stable optimization landscapes, we conducted a comprehensive structural and hyperparameter search. Optuna was utilized to explore learning rates (sampled log-uniformly between 10^{-6} and 10^{-2}), latent dimensionalities ($\{32, 64, 128, 256\}$), optimizer selections (Adam vs. RMSprop), and the Kullback-Leibler regularization weight

($\beta \in [0.1, 2.0]$). For the dVAE, the input noise corruption factor was additionally tuned between 0.05 and 0.30. Each trial was evaluated using early stopping with a MedianPruner, terminating unpromising runs early to maximize computational efficiency.

In contrast, GANs and Diffusion Models exhibit highly sensitive training dynamics and are prone to collapse if core parameters deviate from theoretical bounds. Therefore, rather than conducting blind grid searches, we anchored our structural choices and optimizer moments to established literature and course recommendations. For example, adversarial optimizers utilized standard, proven momentum values ($\beta_1 = 0.5, \beta_2 = 0.999$ for DCGAN; $\beta_1 = 0.0, \beta_2 = 0.99$ for WGAN-GP), and the Diffusion models relied on a fixed cosine variance schedule. Optuna was subsequently deployed in a targeted capacity to fine-tune the learning rates and weight decay parameters to optimally map to the specific ArtBench-10 data distribution.

Once the optimal configurations were identified for each model family, the best-performing architectures were retrained from scratch on the full 50,000-image dataset for the final comparative analysis.

4.2 Evaluation Protocol

To address the lack of a single ground-truth output in generative tasks, we evaluated our models using distribution-level metrics: the Fréchet Inception Distance (FID) and the Kernel Inception Distance (KID).

Evaluation Metrics FID compares real and generated samples by projecting them into a pretrained Inception feature space and modeling both distributions as multivariate Gaussians [17]. In contrast, KID functions similarly by extracting features, but it utilizes Maximum Mean Discrepancy (MMD) to measure the distance between distributions in a reproducing kernel Hilbert space , rather than assuming a Gaussian distribution [17].

Implementation Details These metrics measure the distance between the feature representations of real and generated images extracted from a pre-trained Inception-v3 network, as implemented in the TorchMetrics library. For the final evaluation of each model, we generated a set of $N = 5,000$ synthetic images and compared them against $5,000$ real images sampled from the ArtBench-10 test split. To compute the KID, we utilized 50 random subsets of size 100.

Statistical Robustness Generative models, particularly GANs, can exhibit high variance depending on the random noise vectors sampled during inference. To guarantee the statistical reliability of our results, we repeated the evaluation protocol 10 times for each trained model. For each of the 10 iterations, we enforced strict deterministic behavior by updating the pseudo-random number generator seeds (`numpy` and `torch` seeds initialized to $100 + i$, where i is the run index). We report the final FID and KID scores as the Mean $\pm$ Standard Deviation across these 10 independent runs.

5 Analysis of the Results

5.1 Quantitative Results

As shown in Table 1, the dVAE model outperformed all other VAE variants tested, with the following optimal hyperparameter configuration: learning rate (lr) = 0.0014, latent dimension (d_{latent}) = 128, β = 0.1099, and noise factor = 0.1713. The dVAE's superior validation loss demonstrates that noise injection acts as an effective regularizer, compelling the encoder to learn robust structural representations rather than surface-level details. Furthermore, it is also worth noting that ConvVAE2, despite its greater capacity, did not substantially outperform the base ConvVAE in terms of final quality.

Table 1: Performance comparison of various VAE architectures and their optimal hyperparameters.

Model	lr	**Latent Dim**	**Opt.**	**Loss**	β	**Noise**	**Best Loss**
Conv VAE	0.0001	256	Adam	MSE	1.2227	–	75.1357
Conv VAE2	0.0027	64	Adam	MSE	0.4354	–	60.0437
Denoising VAE	0.0014	128	Adam	MSE	0.1099	0.1713	37.1941

Within the Generative Adversarial Network (GAN) family, we evaluated the impact of advanced stabilization techniques by comparing our baseline DCGAN against the WGAN-GP. As detailed in Section 3, standard GANs utilizing the Jensen-Shannon divergence are prone to vanishing gradients and mode collapse. The WGAN-GP mitigates this by optimizing the Wasserstein-1 distance and enforcing a 1-Lipschitz constraint via a gradient penalty.

Table 2: Intra-family performance comparison of the trained GAN architectures.

Model	**Loss Function**	**Normalization (Critic)**	**FID (↓)**	**KID (↓)**
DCGAN	Hinge Loss	Spectral Norm	48.21 ± 0.20	0.0269 ± 0.0007
WGAN-GP	Wasserstein + GP	Layer Norm	43.07 ± 0.44	0.0230 ± 0.0009

The results in Table 2 confirm the theoretical advantages of the Wasserstein framework. The WGAN-GP successfully lowered the FID from 48.21 to 43.07. By replacing Spectral Normalization with Layer Normalization and utilizing the gradient penalty, the WGAN-GP maintained healthier gradient flow throughout training, resulting in a more robust mapping of the ArtBench data distribution and a measurable improvement in generation quality.

Finally, within the Diffusion family, we explored the trade-off between absolute visual fidelity and computational efficiency by comparing a standard Pixel-Space Diffusion model against a Latent Diffusion Model (LDM).

Table 3: Intra-family performance comparison of the trained Diffusion architectures.

Model	**Operating Space**	**Spatial Resolution**	**FID (↓)**	**KID (↓)**
Pixel Diffusion	RGB Pixel Space	$32 \times 32 \times 3$	24.08 ± 0.19	0.0082 ± 0.0003
Latent Diffusion	Latent Space	$8 \times 8 \times 4$	101.56 ± 0.88	0.0641 ± 0.0014

As shown in Table 3, the Pixel Diffusion model vastly outperformed the LDM. This stark contrast (FID 24.08 vs 101.56) is highly indicative of a perceptual compression bottleneck. While the LDM benefits from operating in a heavily compressed $8 \times 8 \times 4$ space, making the reverse diffusion process computationally lightweight, its maximum generation quality is strictly bounded by the decoding capabilities of the frozen ConvVAE. Because the Autoencoder naturally penalizes high-frequency details (resulting in blurrier reconstructions), the diffusion network is structurally incapable of synthesizing the sharp, intricate brushstrokes that the Pixel-Space model can easily generate.

Overall Comparative Analysis As mandated by our development strategy, we selected the best-performing architecture from each of the three model families to be retrained on the full 50,000 image dataset and subjected to our robust evaluation protocol. Table 4 presents this final comparative evaluation across the three generative paradigms.

Table 4: Final quantitative evaluation of the best-performing models from each family on the full ArtBench-10 dataset (Mean ± Std across 10 random seeds).

Model Family Winner	**FID (↓)**	**KID (↓)**
dVAE (Autoencoders)	162.32 ± 0.64	0.1700 ± 0.0012
WGAN-GP (GANs)	43.0661 ± 0.4466	0.0230 ± 0.0009
Pixel Diffusion (Diffusion)	24.0786 ± 0.1921	0.0082 ± 0.0003

As shown in Table 4, the performance differences between the generative paradigms are stark. The Pixel-Space Diffusion model achieved the absolute best distribution-level metrics, with an FID of 24.08 and a KID of 0.0082, significantly outperforming the other families. This confirms recent literature indicating that iterative denoising excels at mapping complex, multi-modal distributions—such as the varied brushstrokes and color palettes of different artistic styles—without suffering from mode collapse.

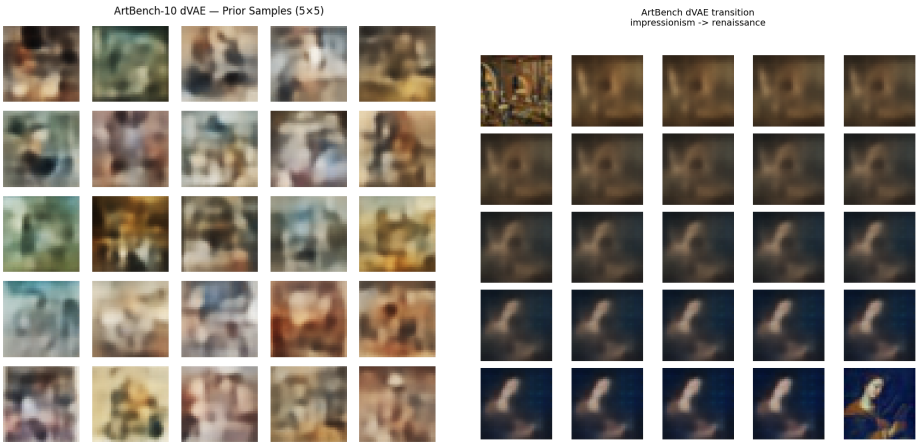
The adversarial representative, WGAN-GP, positioned itself as a strong intermediate solution. With an FID of 43.07, it lagged behind the diffusion model in absolute fidelity but maintained a massive lead over the VAE framework. This places GANs in a favorable position for scenarios where the high computational cost and slow inference speed of diffusion models are prohibitive, but visual sharpness is still required.

Finally, the dVAE exhibited the lowest generative performance (FID: 158.01). While expected, as the model's reliance on pixel-wise reconstruction constraints tends to induce excessive smoothing and a loss of fine-grained structural details, our evaluation confirms that autoencoder-based approaches struggle to match the perceptual quality of GANs and diffusion models.

5.2 Qualitative Results

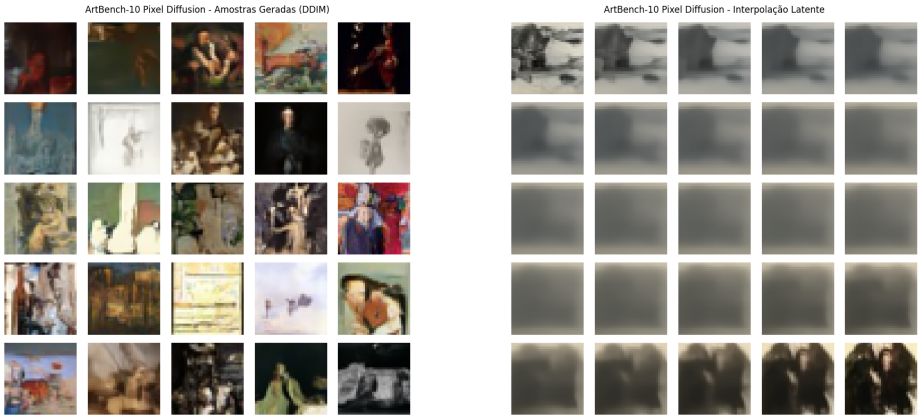
To corroborate our quantitative metrics, we conducted a visual inspection of the generated samples and latent space traversals. Figure 2 illustrates the outputs of our best-performing model, the Pixel-Space Diffusion.

As reflected by the superior FID score, the Diffusion samples exhibit fine granular details, coherent artistic structures, and high stylistic diversity without obvious signs of mode collapse. Furthermore, the latent space interpolations (Figure 2b) demonstrate smooth, continuous semantic transitions between distinct artistic styles, confirming that the model learned a well-structured geometric manifold rather than simply memorizing the training data.



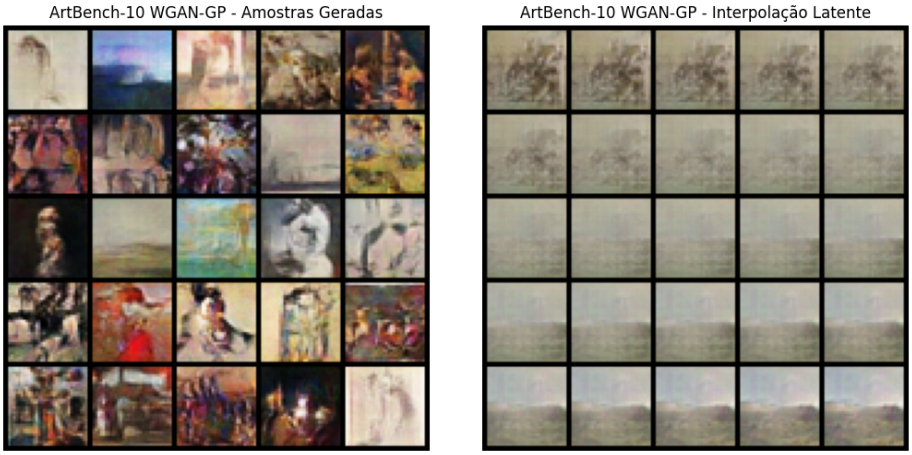
a) Random Samples b) Latent Space Interpolation

Fig. 1: Qualitative results for the Denoising VAE model. Random sampling (a) preserves global structure but sacrifices fine detail, whereas latent interpolation (b) offers smooth transitions with inconsistent fidelity.



a) Random Samples b) Latent Space Interpolation

Fig. 2: Qualitative results for the Pixel Diffusion model. The random samples (a) exhibit high structural fidelity and diversity. The latent interpolation (b) demonstrates smooth transitions between artistic styles, confirming a well-structured latent space.



a) Random Samples b) Latent Space Interpolation

Fig. 3: Qualitative results for the WGAN-GP model. While the generated samples (a) capture broad stylistic colors, they exhibit more visual artifacts compared to the diffusion model. The latent interpolation (b) shows the model's ability to transition between concepts, though with slightly less structural coherence.

6 Conclusion

In this project, we successfully implemented and evaluated three distinct families of generative models: VAEs , GANs, and Diffusion Models on the ArtBench-10 dataset. Our comparative analysis revealed clear trade-offs between architectural complexity, training stability, inference speed, and ultimate sample quality.

The Pixel-Space Diffusion model emerged as the definitively superior architecture in terms of distribution-level metrics, achieving an FID of 24.08. It demonstrated an exceptional ability to capture the complex, multi-modal distribution of artistic styles without suffering from mode collapse. However, this quality came at the cost of significantly slower inference times compared to single-pass models.

In contrast, the dVAE (with an FID of 162.32 and a KID of 0.1700) represents the lower end of the generative quality spectrum. While it is the weakest model in terms of perceptual fidelity, this outcome is expected and highlights the inherent design of the VAE framework. Nonetheless, the dVAE serves as a critical baseline. Its superior validation loss compared to the ConvVAE variants suggests that noise injection effectively regularizes the encoder, forcing it to capture structural representations rather than memorizing surface details. Thus, while it does not compete with Diffusion or GANs in visual fidelity, it provides essential insights into the bottleneck of latent-variable models.

Within the adversarial family, we observed that while the baseline DCGAN was prone to instability, the implementation of the Wasserstein distance and gradient penalty (WGAN-GP) successfully stabilized training and improved the FID to 43.07. GANs remain highly competitive when inference speed is prioritized over absolute distributional perfection.

Finally, our exploration of Latent Diffusion Models (LDM) highlighted the critical importance of the compression stage. The high FID (101.56) of our LDM indicates a perceptual compression bottleneck; the diffusion process is strictly bounded by the detail-retention capabilities of the underlying Autoencoder.

Despite these findings, this study is subject to certain limitations that open avenues for future work. Due to excessive computational requirements, we were unable to conduct an exhaustive hyperparameter optimization, such as varying batch sizes or testing alternative variance schedules for Diffusion models, which could have potentially improved model performance. Furthermore, while we relied on FID and KID as standard benchmarks, recent literature suggests that these metrics may not always fully align with human perceptual judgments [1].

Ultimately, this study confirms that while Diffusion models currently represent the state-of-the-art for high-fidelity artistic synthesis, architectural selection must be guided by the specific constraints of the deployment environment, balancing the need for visual quality against computational overhead.

Acknowledgments

During the development of this project, we utilized Generative AI tools to assist with code debugging and hyperparameter tuning strategies with Optuna. All AI-assisted components were reviewed and adapted to ensure correctness, originality, and alignment with project goals.

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